Logarithms

└ are a way of looking at Exponentials

eg: 2^5 where: 2 is the base

"an expansion" 5 is the exponent/index/power

base is like growth rate

exponent is like time

$2^0 = 1$ like 'no growth' so result stay same (100%) no change. No time growing

$2^{-1} = \frac{1}{2}$ 'negative growth'

$\frac{1}{2}$ b/c you are in a doubling machine (function)

eg continued in log
└ converted to } 'Sort of shuffling of terms to solved for time'

$\log_2 32 = 5$ "log base 2 of 32 equals 5"

$\log_2 1 = 0$ "at growth rate of 2 and a result of 1 means we grew for 0 time"

$\log_2 \left(\frac{1}{2}\right) = -1$

eg $\log_3 81 = 4$

└ 4 rotations in the tripling machine $3^4 = 81$

$\log_{\sqrt{3}} 27 =$

$\sqrt{3} \cdot \sqrt{3} = 3$ ... so... $\sqrt{3}^n = 27$ $n = ?$